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Overprint Magnetizations in Support of Northward Displacement of the Chugach-Prince William Terrane, Alaska

ALAN J. BOL[1]

Geophysics Department, Stanford Univeristy, Stanford, California

Widespread discrepancies between geologic and paleomagnetic estimates of northward displacement are ubiquitous in reconstructions of the North American Cordillera. In southern Alaska, this discrepancy persists into Paleogene time. Paleomagnetic data indicate that the Mesozoic arc complex of the Southern Alaska terrane was in place by Late Cretaceous-early Tertiary time, while the outboard Mesozoic and Tertiary accretionary complex of the Chugach-Prince William terrane still lay far to the south. The apparent Tertiary displacement of the Chugach-Prince William terrane is difficult to reconcile with most apparent geologic constraints. I studied the paleomagnetism of early Tertiary mafic volcanic rocks from Prince William Sound, Alaska, in an attempt to corroborate the meager paleomagnetic database. I sampled at Knight Island and Glacier Island and found overprint magnetizations. At Knight Island, the paleomagnetic directions are too scattered for tectonic interpretation. At Glacier Island, I found three overprint magnetizations. The type 1 magnetization has a normal direction and is concordant with the expected direction for North America. The type 2N magnetization yields a characteristic direction which is too scattered for tectonic interpretation. The type 2R magnetization is reversed and implies 17°±10° of northward displacement with respect to North America assuming that the magnetization was acquired at 55 Ma and the ophiolite was not subsequently tilted. The type 2R magnetization agrees remarkably well with paleomagnetic results from the nearby Resurrection Peninsula in terms of inferred displacement, paleotrend of the sheeted dike complex, and reversed polarity of magnetization. The type 1 magnetization was subsequently acquired after the Chugach-Prince William terrane arrived in southern Alaska.

INTRODUCTION

Perhaps the most vexing problem to plague Cordilleran tectonics is the discordance between terrane displacement inferred from paleomagnetic studies and that inferred from geologic studies [e.g., *Beck*, 1991; *Gastil*, 1991]. Large poleward displacements indicated by paleomagnetic studies remain unaccounted for in the currently accepted displacement of major faults and remain unnecessary in most geologically based reconstructions. However, many geologic units share strong similarities (e.g., Jura-Cretaceous arc/forearc sequences) and structural features are commonly orogen parallel. Therefore distinct pierce points on faults are often lacking, and large displacements remain allowable.

In southern Alaska, the most reliable paleomagnetic studies of lower Tertiary rocks from the Chugach-Prince William terrane (CPW, Figure 1)[*Plafker et al.*, 1977, 1989; *Bol and Gibbons*, 1992] indicate significant poleward displacement of the terrane (Figure 2*a*). These studies were composed of a large number of samples, passed stability tests, and used volcanic rocks which are not susceptible to compaction-induced inclination shallowing. This large displacement is not indicated by the most reliable paleomagnetic studies of similar aged rocks from the more inboard Southern Alaska terrane (composed of the Alexander, Wrangellia, and Peninsular terranes, Figures 1 and 2b)[*Harbert*, 1990]. Geologic studies support the hypothesis that the Southern Alaska terrane arrived at or near its present position by Late Cretaceous or early Tertiary time [e.g., *Csejtey et al.*, 1982; *Nokleberg et al.*, 1985; *Wallace et al.*, 1989; *Davies and Moore*, 1984]. This places the CPW at or near its present position in early Tertiary time, contradicting the paleomagnetic studies. Furthermore, no faults are known to possess the requisite displacement to account for the translation of the CPW with respect to the Southern Alaska terrane [*Plafker*, 1987].

The paleomagnetic database for the CPW is small (six studies) and reliability and inferred displacements vary widely. Most results are from pillow lavas, whose paleomagnetism is often poorly understood and susceptible to overprints. The two most reliable studies, from the Ghost Rocks Formation [*Plumley et al.*, 1983] and the Resurrection Peninsula ophiolite [*Bol et al.*, 1992](Figure 1), indicate consistent, significant, poleward displacement. Recent geologic studies support this hypothesis and imply several hundred kilometers of cumulative displacement across the CPW (discussed below).

In Prince William Sound, paleomagnetic studies of lower Tertiary volcanic and ophiolitic rocks have yielded questionable and contradictory results. A reconnaissance study of the Knight Island ophiolite by *Hillhouse and Grommé* [1977] yielded equivocal results that required no displacement with respect to cratonal North America. This is the only study in the CPW that yields a paleomagnetic pole concordant with the North American pole. However, the data comprise a small number of stably magnetized sites (n=9 of 15 sites sampled), the scatter was large, and the tilt correction was uncertain. Despite these limitations, the remanence passed a fold test and both polarities of magnetization were present. Paleomagnetic studies of apparently correlative rocks from Galena Bay and Wingham Island put the CPW at near-equatorial paleolatitudes [*Plumley and Plafker*, 1985] (P. W. Plumley, written communication, 1989). The remanent magnetization of these rocks are also reported to pass fold tests. Unfortunately, these results have been published only in abstract form, so their reliability is difficult to evaluate.

In this paper I report the results of a paleomagnetic investigation of the volcanic rocks exposed on Knight Island

[1]Now at Boulder, Colorado.

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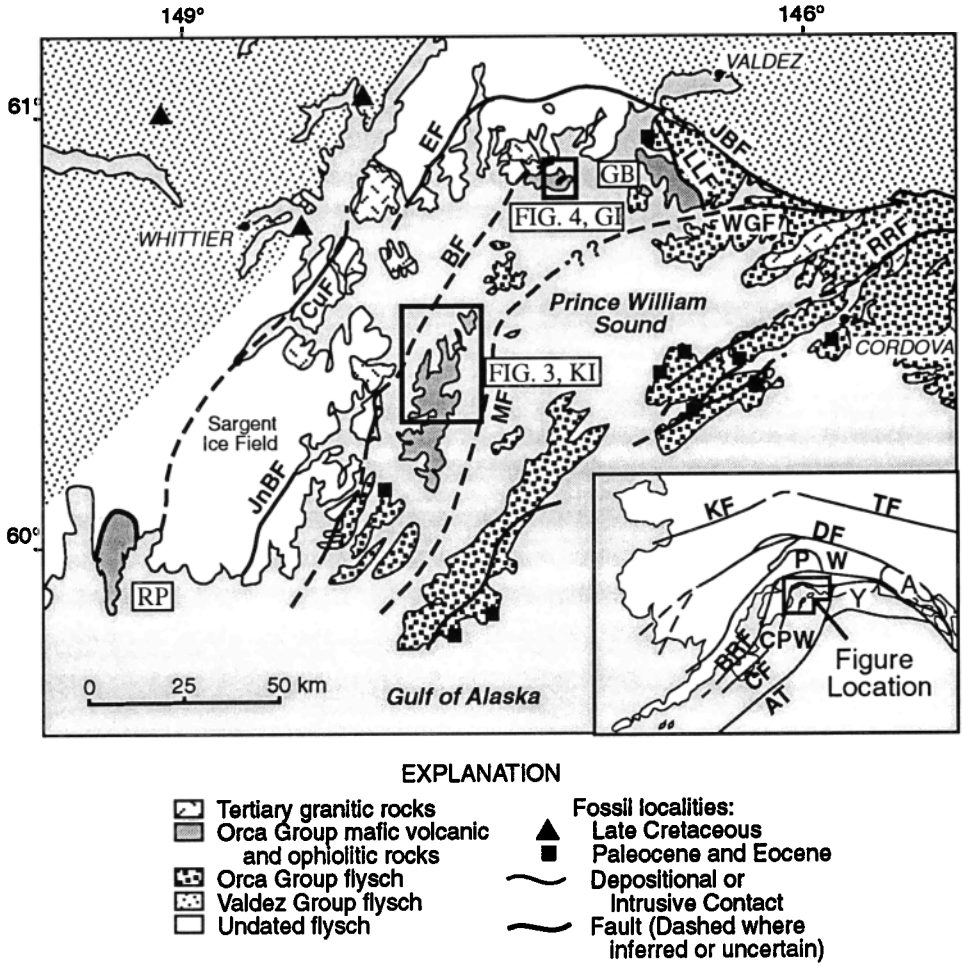


Fig. 1. Generalized geologic map of the Prince William Sound region [after *Tysdal and Case*, 1979; *Winkler and Plafker*, 1981; *Winkler et al.*, 1981; *Nelson et al.*, 1985] with locations of Figure 3 (KI, Knight Island) and Figure 4 (GI, Glacier Island). Inset map shows major terranes and faults of south central Alaska [from *Coney et al.*, 1980; *Jones et al.*, 1983] as well as location of Prince William Sound. Abbreviations are A, Alexander terrane; AT, Aleutian Trench; BF, Bainbridge fault; BRF, Border Ranges fault; CPW, Chugach-Prince William terrane; CF, Contact fault; CuF, Culross fault; DF, Denali fault; EF, Eaglek fault; GB, Galena Bay; GI, Glacier Island; JBF, Jack Bay fault; JnBF, Johnstone Bay fault; KI, Knight Island; KF, Kaltag fault; LLF, Landlock fault; MF, Montague fault; P, Peninsular terrane; RP, Resurrection Peninsula; RRF, Rude River fault; TF, Tintina fault; W, Wrangellia; WGF, Western Gravina fault; and Y, Yakutat terrane.

and Glacier Island in Prince William Sound (Figure 1). This study revealed only overprint magnetizations. At Knight Island, the data are scattered and cannot be interpreted tectonically; therefore the previous reconnaissance result should be abandoned. At Glacier Island, the overprint directions are well grouped, and my preferred interpretation is that of two overprints: a reversed direction acquired soon after formation at a location near present-day Washington or Vancouver Island, followed by a normal overprint acquired after arrival of the CPW in southern Alaska.

Below, I summarize the tectonic and geologic setting of the CPW in early Tertiary time and follow this with details of the studies. Last, I interpret the results and discuss the broader implications of the research.

Tectonic and Geologic Setting

Terrane Tectonics of Southern Alaska

Alaska is composed of a collage of lithotectonic terranes, many of them exotic [e.g., *Coney et al.*, 1980]. In southern Alaska, there are two major terranes: the composite Southern Alaska terrane and the CPW (Figure 1). From Jurassic into early Tertiary time, the Southern Alaska terrane was thought to have composed a magmatic arc along the North America continental margin [*Jones et al.*, 1983]. Outboard of the Southern Alaska terrane lies an accretionary prism composed dominantly of Upper Cretaceous and Lower Tertiary flysch. The Mesozoic portion of the accretionary prism constitutes the Chugach terrane, and the Tertiary portion of the accretionary prism constitutes the Prince William terrane [*Plafker et al.*, 1977]. Recent work has focused on the difficulty of distinguishing the Chugach and Prince William terranes, and many researchers now consider them a long-lived accretionary prism, best grouped together in a single Chugach-Prince William terrane [*Dumoulin*, 1988; *Bol and Gibbons*, 1992].

The most reliable paleomagnetic results indicate that the CPW originated 13° to 25° to the south of its present position [*Plumley et al.*, 1983; *Bol et al.*, 1992]. Approximately 5° of the northward movement can be accommodated with geologically inferred displacement along the inboard Denali fault and the Tintina fault/Northern Rocky Mountain Trench

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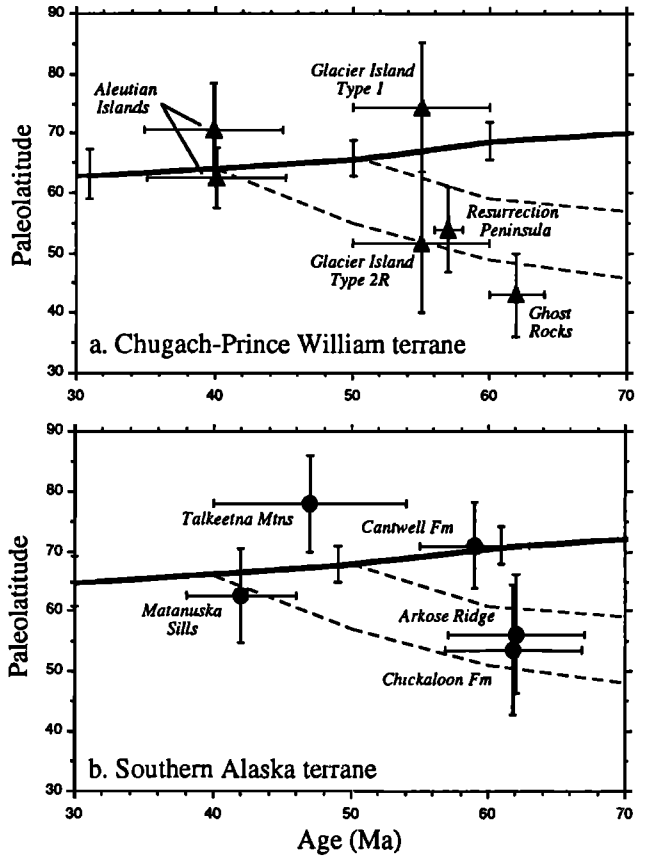


Fig. 2. Plots of paleolatitude versus age for the (a) Chugach-Prince William terrane and (b) Southern Alaska terrane. In both plots, the thick solid line is the expected paleolatitude for a point in the eastern Chugach mountains fixed to cratonal North America [*Diehl et al.*, 1983] and the thin dashed line is the trajectory of a point in the eastern Chugach mountains moving with the full coastwise displacement of the Kula plate and arriving at either 40 or 50 Ma [*Debiche et al.*, 1987]. In Figure 2*a*, studies from the Resurrection Peninsula ophiolite [*Bol et al.*, 1992] and the Ghost Rocks formation [*Plumley et al.*, 1983] indicate significant poleward displacement in Eocene time with an arrival time of about 40 Ma, confirmed by results from the Aleutian Islands [*Harbert et al.*, 1987]. The results from Glacier Island are the subject of the present study and are discussed in detail later in the text. In Figure 2*b*, results from the Southern Alaska terrane are generally in place with respect to cratonal North America in Paleogene time, with the exception of results from the Chickaloon and Arkose Ridge formations. The latter two studies are from sedimentary rocks and may be subject to inclination flattening, while the nearly coeval result from the Cantwell formation comes from volcanic rocks and is probably more reliable. References for Figure 2*b* are as follows: Arkose Ridge and Chickaloon formations [*Stamatakos et al.*, 1989], Cantwell formation [*Panuska and Macicak*, 1986; *Hillhouse and Grommé*, 1982], Matanuska sills [*Stamatakos et al.*, 1988], Talkeetna Mountains [*Panuska et al.*, 1990].

(Figure 1)[e.g., *Roddick*, 1967; *Gabrielse*, 1985; *Nokleberg et al.*, 1985]. However, geologic and paleomagnetic studies from the Southern Alaska terrane, immediately inboard of the CPW, indicate little or no movement with respect to cratonal North America since Late Cretaceous-early Tertiary time [e.g., *Wallace et al.*, 1989; *Csejtey et al.*, 1982; *Coe et al.*, 1989; *Panuska et al.*, 1990]. Thus right-lateral displacement appears to be required between the inboard (Southern Alaska) and outboard (CPW) terranes.

Recent studies suggest that the Border Ranges fault system and the Contact fault system may have accommodated large dextral displacements. Apparent offset provenance across the Border Ranges fault system suggests >100 km of post-Cretaceous dextral slip [*Nelson et al.*, 1986] and structural relations suggest at least a few tens of kilometers of Tertiary dextral offset [*Little*, 1990; *Roeske et al.*, 1991]. This displacement occurs between the Southern Alaska terrane and the CPW and helps account for the discrepancy in their respective displacements. Across the Contact fault system, a significant crustal discontinuity was observed in seismic refraction studies of the Trans-Alaska Crustal Transect of the U.S. Geological Survey [*Fuis et al.*, 1991; *Wolf et al.*, 1991]. In conjunction with field evidence, *Bol and Roeske* [1993] interpreted this discontinuity to indicate at least several tens, if not hundreds, of kilometers of dextral offset along the Contact fault system. Sedimentologic, isotopic, and petrographic studies suggest a provenance for the CPW in the Coast Mountains of British Columbia, several hundred kilometers to the south [*Hollister*, 1979; *Farmer et al.*, 1989; *Sample and Reid*, 1993], in agreement with the paleomagnetic results.

Contradicting the hypothesis of large displacement, plutons of the Sanak-Baranof magmatic belt [*Hudson et al.*, 1979] intrude both the CPW and Southern Alaska terrane on Kodiak Island and apparently stitch the inboard and outboard terranes together by Paleocene time [*Davies and Moore*, 1984]. This relation led *Thrupp and Coe* [1986] to hypothesize a broad zone of distributed dextral shear between inboard and outboard terranes to accommodate the relative displacement without disrupting this loose geologic link. Supporting this hypothesis, dextral-slip faulting has been observed in the northern Chugach Mountains [*Little*, 1990], on the Kenai Peninsula [*Mitchell*, 1979; *Bradley and Kusky*, 1991], and along the Contact fault system in eastern Prince William Sound [*Bol and Roeske*, 1993] and in the Kodiak Islands [*Sample and Moore*, 1987]. In summary, the case for or against large northward translation of the CPW remains unresolved; our only hope is to gather more data until an overwhelming preponderance favors one hypothesis over the other.

Geology of the Chugach-Prince William Terrane

The rocks of the CPW dominantly are variably deformed and metamorphosed flysch, with subordinate mafic volcanic rocks, and two suites of plutonic rocks [e.g., *Plafker et al.*, 1977, 1989; *Moore et al.*, 1983; *Nelson et al.*, 1985]. These rocks range in age from Triassic through middle Eocene [*Jones and Clark*, 1973; *Plafker et al.*, 1985; *Blome et al.*, 1990], generally younging seaward. Metamorphism and deformation intensity generally decrease seaward, although the pattern is locally complex [*Miller et al.*, 1984; *Plafker et al.*, 1989; *Bol and Gibbons*, 1992].

The mafic volcanic rocks and ophiolite sequences of the CPW constitute a volumetrically minor, but important rock unit. Early Tertiary mafic volcanic rocks are especially abundant. These rocks are believed to have originated in two ways; an inboard suite is related to subduction of the extinct Kula-Farallon ridge [*Hill et al.*, 1981; *Moore et al.*, 1983], and an outboard suite is believed to have originated in a different fashion, perhaps at seamounts or along a "leaky transform" (G. Winkler, written communication, 1989; S. Nelson, personal communication, 1991). In the region of Prince William Sound, rocks of the inboard MORB-like suite include those studied for paleomagnetism at the Resurrection Peninsula, Knight Island, Glacier Island, and Galena Bay. Geologic, petrographic, and geophysical evidence suggest that these rocks are correlative: (1) the Resurrection Peninsula, Knight Island, and Glacier

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Island all have a stratigraphy of sheeted dikes overlain by pillow basalt interbedded with flysch [*Tysdal et al.*, 1977]; (2) the flysch interbedded with pillow basalt at the Resurrection Peninsula, Knight Island, and Galena Bay has an anomalously quartzofeldspathic composition compared to the surrounding flysch [*Dumoulin*, 1988]; and (3) a prominent gravity and aeromagnetic anomaly appears to connect Knight Island, Glacier Island, and Galena Bay in Prince William Sound [*Nelson et al.*, 1984].

Because the apparent polar wander for cratonal North America was unusually fast in Late Cretaceous and early Tertiary time, the poleward displacement indicated by paleomagnetic studies is very sensitive to the age of the rocks. Unfortunately, rocks of the CPW are generally very poorly dated. Fossils are scarce, poorly preserved, and frequently do not yield precise age dates [e.g., *Jones and Clark*, 1973; *Plafker et al.*, 1985]. Isotopic age dating has proven to be effective only on plutonic and metamorphic rocks [e.g., *Lanphere*, 1966; *Sisson et al.*, 1989]; with few exceptions, mafic volcanic rocks remain undated. However, *Nelson et al.* [1989] recently obtained a U-Pb (zircon) date of 57±1 Ma from plagiogranite of the Resurrection Peninsula ophiolite. The plagiogranite intrudes the sheeted dike complex and is in turn cut by dikes, thus it dates the emplacement of the ophiolite.

Knight Island. The geology of Knight Island is composed of the upper portions of an ophiolite exposed in a shallowly north plunging anticlinorium with a steeply dipping east flank and a moderately dipping west flank (Figure 3). The sheeted diabase dike complex is exposed along the rugged spine of the island, and the overlying pillow basalt section is found along the margins. The pillow basalt section appears to wrap around the north end of the island, defining the plunging fold. Pillow basalts on the east flank of the structure are highly disrupted and deformed, while those on the west flank appear undeformed and relatively unmetamorphosed. Sequences up to several meters thick composed of thin silty turbidites are commonly found in the upper portions of the pillow basalt section. Diabase dikes are found throughout the pillow basalt section but are more common in the lower portions.

Glacier Island. The rocks of Glacier Island are similar to those of Knight Island: sheeted dikes, pillow basalt, and minor sequences of thin, silty turbidites (Figure 4). The pillows mostly dip to the northwest, with one area where they dip back to the southeast. The map pattern combined with the stereonet plot of poles to bedding indicates that the overall pattern is a north-northeast trending syncline. The plot of poles to dikes does not show a girdle like the plot of poles to bedding, most likely because the dikes are nearly perpendicular to the best fit regional fold axis. Dikes and pillows were not mutually perpendicular at many localities, but dikes can be observed feeding and grading into pillows along the south coast of the island; thus dikes and pillows appear coeval. Dikes intruding the pillows are much more common at Glacier Island that at Knight Island.

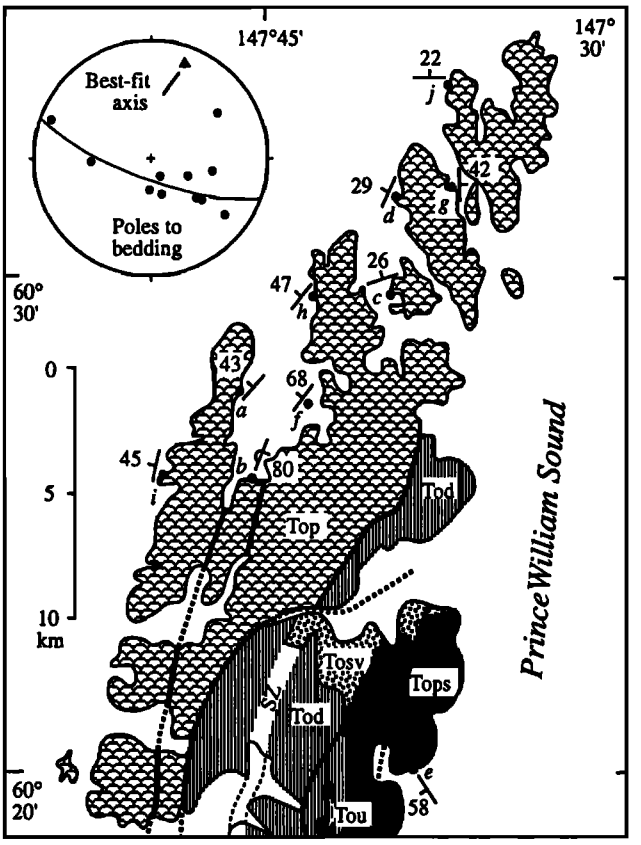


Fig. 3. Generalized geologic map of northern Knight Island [from *Tysdal and Case*, 1979; *Nelson et al.*, 1985] with localities and their respective bedding attitudes. Localities Ka-Kj are labeled a-j. Equal-area plot in upper left corner, combined with map pattern, indicates that the overall structure is a gently north plunging anticline. Map abbreviations are Top, pillow basalt with minor interbedded, thin, silty turbidites; Tod, sheeted diabase dikes; Tops, pillow basalt with subordinate sedimentary rocks; Tosv, sedimentary rocks with subordinate volcanic rocks; and Tou, ultramafic rocks.

PALEOMAGNETIC RESULTS

Field and Laboratory Techniques

I collected samples from fresh, coastal exposures that offered good bedding: from either interbedded sediments, well-layered basalt pillows, or basalt flow tops. Three to 16 bedding attitudes were collected at each site (usually 5-10; Table 1*a*), with fewer attitudes at sites with sedimentary bedding. At each location where I thought bedding could be accurately determined, I typically collected six samples from each of three sites. Samples were 2.5-cm diameter cores taken with a portable drill and oriented with a magnetic compass. Small, local declination anomalies were corrected by backsighting to known landmarks. In the laboratory, 2.5-cm specimens were cut from the cores. Companion pilot specimens from one sample per site were treated with detailed stepwise thermal and alternating field demagnetization. The remainder of the samples were thermally demagnetized over 10-14 steps.

Data were analyzed by first studying orthogonal vector diagrams [*Zijderveld*, 1967]. Remanence directions from samples that displayed a stable endpoint magnetization were determined by a least squares fit to three or more steps [*Kirschvink*, 1980]. Samples whose remanence direction curved into or "missed" the origin on the orthogonal vector diagrams were analyzed by fitting planes to the data [*Kirschvink*, 1980]. Data from stable endpoints (and planes, where appropriate) were combined into site mean directions [*McFadden and McElhinney*, 1988].

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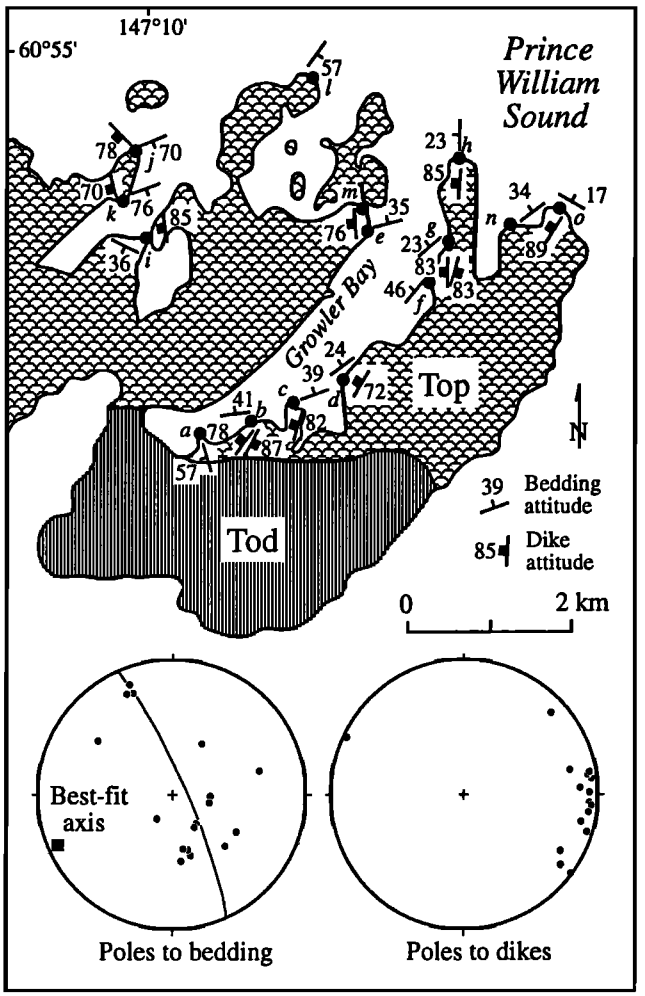


Fig. 4. Generalized geologic map of eastern Glacier Island [from *Tysdal and Case*, 1979; *Nelson et al.*, 1985] with localities and their respective bedding and dike attitudes. Localities Ga-Go are labeled a-o. Equal-area plots of poles to bedding and poles to dikes are shown at the bottom of the figure. Plot of poles to bedding combined with map pattern indicates that the overall pattern is a south-southwest trending syncline. Map abbreviations are Top, pillow basalt with minor interbedded, thin, silty turbidites; and Tod, sheeted diabase dikes.

Knight Island

Sampling. At Knight Island , I collected 255 cores from 43 sites (Figure 3). The variation in bedding attitudes (Figure 3 and Table 1a) is sufficient to produce a conclusive fold test. That is, the variation in tilt far exceeds any that might result from initial topography ("original dip").

Demagnetization behavior. The stability of the remanent magnetization and the degree to which it "cleaned up" during stepwise demagnetization varied considerably from locality to locality but was fairly consistent within localities. A total of 17 sites from five localities displayed stable endpoint magnetizations with blocking temperatures near 580°C (Figure 5*a*). I found little difference between the results of thermal and alternating field (af) demagnetizations for these samples (Figure 5*b*). However, in some samples, af demagnetization did not completely remove secondary magnetizations, so I employed thermal demagnetization as a rule.

Samples from several sites which did not show a stable endpoint magnetization revealed a straight-line trajectory between 200° and 500°C (Figure 5*c*). These samples were analyzed by fitting great circles to the demagnetization plane. Planes were then combined with stable endpoint directions to yield site mean directions. Table 2*a* includes a tally of which sites utilized best fit planes in their site means.

Of the four localities where silty sediments were present, three localities (12 sites) failed to yield any stable magnetizations (regardless of sample lithology) and the fourth locality (Kh) exhibited anomalous demagnetization behavior and yielded relatively low precision data. The reason for this is unclear, but I suspect that the fine-grained sediments are a barrier to the normal convection-driven hydrothermal circulation. In a spreading-ridge environment, this may impede the alteration of relatively unstable titanomagnetite to more stable Ti-poor magnetite and Fe-poor ilmenite, which resists further alteration [e.g., *Facey et al.*, 1985]. Magnetic minerals which remain unaffected by hydrothermal metamorphism are subject to processes of low temperature alteration and magnetic overprinting.

TABLE 1*a*. Knight Island Structural Data

Locality	Bedding Strike, Dip, degrees	a_{95}	k	N_V/N_S
Ka	221, 43W	9	28	10/0
Kb	201, 100E	13	23	6/0
Kc	252, 26N	3	290	9/0
Kd	205, 29W	4	216	6/0
	242, 14N	10	43	5/0
Ke	146, 58W			0/1
Kf	217, 68W	7	52	8/0
Kg	357, 42E	6	38	15/0
Kh	219, 47W	7	38	3/8
Ki	192, 45W	14	56	0/3
Kj	271, 22N	2	443	5/5

Localities are referred to in Table 2a and Figure 3; a_{95} is alpha-95 [*Fisher*, 1953] 95% confidence limit; k is dispersion parameter [*Fisher*, 1953]; N_V is number of bedding measurements from volcanic rocks; N_S is number of bedding measurements from clastic sedimentary rocks.

TABLE 1*b*. Glacier Island Structural Data

Locality	Bedding Strike, Dip, degrees	a_{95}	k	N_V/N_S	Dike Strike, Dip, degrees
Ga	165, 57W			1/0	
Gb	262, 41N	6.4	249	3/0	26, 87E
					216, 78W
Gc	253, 39N	9.8	40	6/0	193, 79W
					197, 85W
Gd	236, 24N			2/0	210, 72W
Ge	254, 35N	(see Gm)			177, 76W
Gf	224, 46W	9.7	50	5/0	
Gg	230, 23N	10.3	96	3/0	18, 83E
					184, 83W
Gh	183, 23W	9.5	53	5/0	185, 85W
Gi	121, 36S	8.3	46	7/0	170, 85W
					171, 83W
					173, 85W
Gj	66, 69S	5.0	84	10/0	137, 78W
Gk	69, 76S	12.2	19	8/0	167, 70W
Gl	36, 57E	9.7	40	6/0	
Gm	254, 35N	6.8	52	9/0	179, 83W
Gn	258, 34N	11.8	23	7/0	
Go	302, 17N	4.5	131	8/0	216, 89W

Localities are referred to in Table 2*b* and Figure 4; see Table 1*a* for other footnotes.

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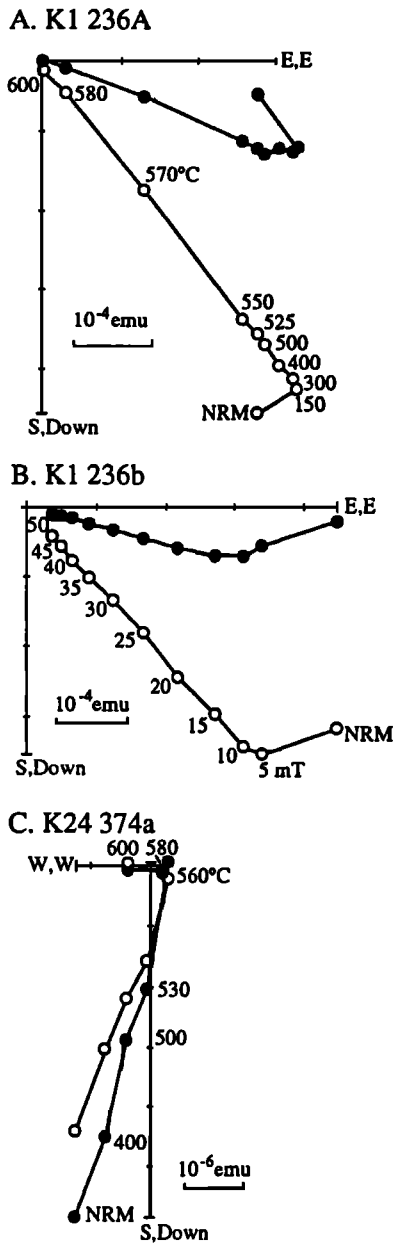


Fig. 5. Orthogonal vector diagrams of samples from Knight Island; open symbols represent the horizontal component and closed symbols represent the vertical component. (*a*) Thermal demagnetization of typical good sample with blocking temperature over 500°C. (*b*) Alternating field demagnetization of a companion specimen to that shown in Figure 5*a*; the directions are nearly identical. However, in less well-behaved samples this was not usually the case. (*c*) Sample whose remanent magnetization did not decay to the origin but "missed." Samples such as this were analyzed by fitting great circles to the demagnetization path.

Rejection criteria. Twelve sites from three locations (11 of them from two locations) did not yield stable remanent magnetizations (Table 2*a*). Two sites had only one or two samples with stable magnetizations--not enough for a reliable mean direction. The remaining sites were screened for dispersion within sites; sites with a *Fisher* [1953] dispersion parameter of $k<10$ were rejected. In this way, six additional sites were rejected.

Results. The site mean data that remain after the rigorous screening above represent only my best data. These data are internally consistent at both the site and locality levels. Unfortunately, they are scattered when taken as a whole, both before and after correction for tilt. The stereonets in Figure 6 show locality mean directions from Knight Island. A careful examination of the locality mean directions reveals that there

TABLE 2*a*. Paleomagnetic Data From Knight Island

Site	Locality	*N/n*	D_g	I_g	D_s	I_s	k	a_{95}	Lith	Status
K1	Ka	7/7	112.2	43.3	49.8	76.3	61.2	7.2	p	
K2	Ka	5/6	121.3	33.2	99.5	74.4	41.5	10.7	f	
K3	Ka	8/8	122.8	44.8	59.1	83.9	56.4	7.0	p	
K4	Kb	7/7	100.9	24.2	307.0	54.4	33.6	9.8	p	
K5	Kb	6/6	88.5	49.9	307.1	27.0	57.7	8.1	p	
K6	Kb	8/8	96.5	40.9	304.8	37.4	90.5	5.5	f	
K7	Kc	7/7	154.4	64.9	53.2	86.6	206.9	3.9	p	
K8	Kc	6/7	139.2	65.3	54.2	80.2	203.8	4.3	p	
K9	Kc	4/5	102.9	-16.1			6.8	32.3	d	k
K10	Kd	6(1)/7	213.7	70.9	262.2	53.4	89.3	4.8	p	
K11	Kd	6(1)/7	254.3	-33.7	232.1	-52.2	45.6	6.7	f	
K11	Kd	5/7	188.9	70.6	254.7	60.4	43.4	10.5	f	
K12	Kd	3(1)/5	8.6	-64.4	60.7	-57.0	41.5	10.0	p	
K13	Ke	0/6							p	u
K14	Kc	4/5	198.3	79.1	281.0	79.9	32.5	14.1	p	
K15	Kc	6/6	172.4	72.9	222.6	83.8	72.5	7.2	p	
K16	Kb	6/6	101.4	38.8	300.8	40.4	123.0	5.5	p	
K17	Kb	5/6	99.7	37.4	303.0	41.4	95.7	7.0	f	
K18	Kb	5/6	104.0	48.4	296.4	31.3	87.8	7.3	f	
K19	Kb	5/6	56.6	43.2	329.4	17.4	9.5	23.2	p	
K20	Kf	4(2)/5	177.1	-13.9	185.5	29.2	50.2	10.0	p	
K21	Kf	5(2)/6	167.3	-4.2	188.7	42.4	12.8	17.2	p	
K21	Kf	6(4)/6	163.6	41.9	258.1	53.3	24.0	12.4	p	
K22	Kf	4(2)/6	162.0	56.7	279.5	47.0	21.5	13.4	p	
K23	Kf	7(3)/7	56.8	56.3	343.6	29.0	5.4	30.4	d	k
K24	Kg	5(1)/6	215.7	44.3	163.5	55.0	8.7	18.5	p	k
K25	Kg	4(2)/6	209.2	20.9	186.5	36.7	6.1	38.4	p	k
K26	Kg	5/6	287.7	83.7	83.3	53.8	11.8	20.6	p	
K27	Kh	4/6	208.0	-17.0	199.2	-4.0	23.8	16.6	p	
K28	Kh	7/7	233.3	-45.9	187.7	-38.2	4.4	28.7	s	k
K29	Kh	4/6	194.5	-28.3	189.8	-2.7	14.2	18.7	f	
K30	Kh	6/7	46.3	75.9	31.9	43.1	3.3	37.6	f	k
K31	Kh	5/6	131.4	44.8	267.3	87.5	4.9	32.1	p	k
K32	Kh	6/6	107.0	34.5	58.2	70.7	5.2	29.1	f	k
K33	Ki	0/6							s	u
K34	Ki	0/6							f	u
K35	Ki	2/6	175.8	-63.7	132.0	-33.3	--	--	p	n
K36	Ki	1/4	96.1	39.7	65.1	83.6	--	--	p	n
K37-K43	Kj	0/36							f,s,p	u
Locality mean data										
	Ka	3/3	118.9	40.5	73.1	79.1	75.3	11.7		
	Kb	7/7	93.2	41.4	307.4	36.1	25.4	11.3		
	Kc	4/4	159.6	71.7	326.2	88.4	49.0	11.5		
	Kd	3/4	196.2	69.0	252.8	57.3	112.6	9.5		
	Ke	0								u
	Kf1	2/4	162.9	49.3	269.0	50.6	--	--		
	Kf2	2/4	172.1	-9.1	187.0	35.8	--	--		
	Kg	1/3	287.7	83.7	83.3	53.3	--	--		
	Kh	2/6	201.3	22.7	194.5	3.4	--	--		
	Ki	0								k, u
	Kj	0								u

Locality: Ka-Kj correspond to a-j in Figure 3; *N* is number of samples used in analysis; parentheses indicate number of samples analyzed by fitting planes to data, otherwise data were analyzed by fitting lines to data; *n* is number of samples measured; for locality mean data *N* is the number of site means used in analysis, *n* is the number of site means measured; D_g is declination, in situ; I_g is inclination, in situ; D_s is declination, tilt corrected; I_g is inclination, tilt corrected; k is dipsersion parameter [*Fisher*, 1953]; a_{95} is alpha-95 95% confidence limit [*Fisher*, 1953]; lith. is lithology: p is pillow basalt, f is massive basalt flow, d is diabase dike, s is siltstone or fine grained sandstone; status refers to either rejection criteria or remanence type: k signifies k less than 10, n signifies that less than 3 samples compose the mean, u signifies unstable remanence in the samples.

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TABLE 2*b*. Paleomagnetic Data From Glacier Island

Site	Locality	*N/n*	D_g	I_g	D_s	I_s	*k*	a_{95}	Lith	Status
G1	Ga	6/7	29.3	78.8	265.6	40.4	372	3.5	p	1
G2	Gb	7/7	311.4	84.3	346.8	44.6	151	4.9	p	1
G3	Gb	6/6	28.8	85.3	356.0	45.2	440	3.2	d	1
G4	Gb	6/6	303.3	82.9	355.6	44.1	2296	1.4	d	1
G5	Gc	7/7	216.1	82.8	333.0	54.9	29.6	11.3	d	2N
G6	Gc	6/6	293.8	79.4	331.9	43.5	896	2.2	d	1
G7	Gc	6/7	304.4	80.8	335.1	43.5	269	4.1	p	1
G8	Gd	7/7	281.6	75.0	308.1	53.8	173	4.6	p	1
G9	Gd	7/7	309.3	79.4	320.6	55.7	1027	1.9	d	1
G10	Gd	7/8	288.9	75.9	311.6	53.7	362	3.2	d	1
G11	Ge	6/6	35.5	49.8	17.3	23.3	15.3	17.7	p	2N
G12	Ge	4/7	256.9	75.3	320.4	50.8	5.2	44.6	d	k
G13	Gf	5/7	129.1	-74.5	132.5	-28.5	65	9.6	d	2R
G14	Gf	6/6	131.5	56.3	320.5	77.6	73	9.0	p	2N
G15	Gg	4/6	88.6	-67.7	112.8	-49.4	20.4	21.2	d	2R
G16	Gg	4/7	330.9	48.4	328.0	25.7	753	3.4	p	2N
G17	Gg	5/7	126.3	-75.0	134.2	-52.2	169	5.9	d	2R
G18	Gg	1/6	353.1	53.0	343.0	32.6	--	--	p	n
G19	Gh	6/6	194.7	85.2	261.5	65.6	113	6.3	p	1
G20a	Gh	3/7	221.7	76.6	253.5	57.0	757	4.5	d	2N
G20b	Gh	3/7	128.6	-53.7	117.4	-33.6	37	20.6	d	2R
G21	Gi	5/11	140.1	-52.2	93.4	-49.2	125	6.9	d	2R
G22	Gi	7/7	9.7	63.2	252.5	75.7	69	7.3	p	1
G23	Gi	6/7	168.1	-66.1	73.8	-66.1	21.6	14.8	d	2R
G24	Gi	5/7	348.5	72.8	238.5	64.3	425	3.7	d	1
G25	Gj	0/7							p	u
G26	Gj	5/7	197.6	74.7	168.6	8.8	2.8	56.9	d	k
G27	Gk	6/6	110.4	-68.1	357.4	-27.6	507	3.0	d	2R
G28	Gk	4/6	307.5	23.9	225.7	58.6	10.6	29.6	p	2N
G29	Gl	4/6	296.7	17.7	275.0	72.6	18.6	21.9	p	2N
G30	Gm	6/6	47.8	66.4	12.2	40.4	54	9.2	p	1
G31	Gm	3/6	121.8	-57.9	140.2	-28.0	27.8	23.8	d	2R
G32	Gm	6/7	350.1	77.4	345.8	42.5	142	5.6	p	2N
G33	Gm	6/6	71.0	-84.8	155.0	-54.9	41.7	10.5	d	2R
G34	Gn	6/7	265.7	73.0	320.9	50.4	63	8.5	p	1
G35	Gn	3/6	60.4	80.9	2.3	52.4	159	9.8	d	2N
G36	Go	4/6	143.9	-52.5	160.5	-43.8	23.5	19.3	p	2R
G37	Go	6/6	134.6	-72.0	170.7	-62.9	182	5.0	d	2R

Locality: Ga-Go correspond to a-o in Figure 4; *N* is number of samples used in analysis; *n* is number of samples measured; D_g is declination, in situ; I_g is inclination, in situ; D_s is declination, tilt corrected; I_g is inclination, tilt corrected; *k* is dipsersion parameter [*Fisher*, 1953]; a_{95} is alpha-95 95% confidence limit [*Fisher*, 1953]; lith. is lithology: p is pillow basalt, d is diabase dike; status refers to either rejection criteria or remanence type: k signifies *k* less than 10, n signifies that less than 3 samples compose the mean, u signifies unstable remanence in the samples, 1, 2R, and 2N are remanence types for samples from Glacier Island.

does not appear to be any sense of convergence of directions when the tilt correction is either applied or removed. In other words, the data are so scattered that even with large differences in bedding dip, a fold test is inconclusive. The structure is apparently simple and an inclination-only fold test is also inconclusive; complications in the tilt correction do not account for the scatter. Good agreement between bedding attitudes collected from sedimentary rocks and pillow basalt rule out large variations in initial dip. I can only conclude that these rocks acquired their magnetization by different means, over a long period of time, and not always parallel to the ambient field. Therefore these rocks are inappropriate for paleomagnetic analysis.

Glacier Island

Sampling. I collected 234 cores from 37 sites (4-7 cores/site) in pillow basalt and diabase dikes that cut the pillows (Figure 4). At each locality where bedding was established, samples were collected from one or two sites each in pillow basalt and diabase dikes. The bedding correction was then applied to both the pillow basalt and dike samples during analysis.

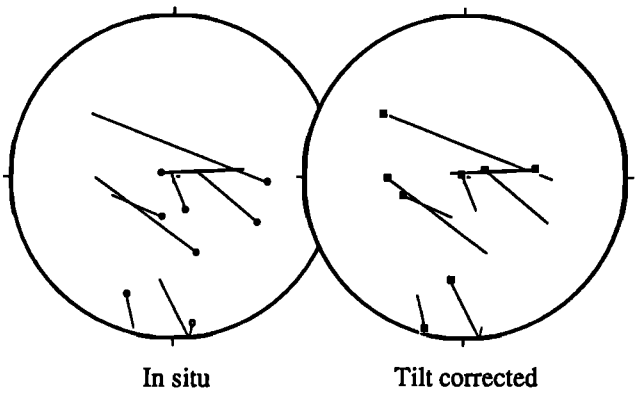


Fig. 6. Equal-area plot of locality mean directions from Knight Island. Lines show the effect of untilting bedding upon directions. While the clustering of directions within sites and within localities is good (Table 2*a*), locality mean directions are scattered both before and after the tilt correction. The lines show that there does not appear to be any convergence of directions when the tilt correction is applied.

Demagnetization behavior. Upon stepwise thermal demagnetization (12-14 steps), the large majority of samples were very stable and revealed straightforward demagnetization paths. Two sites were excluded due to unstable magnetizations and two more sites were excluded due to large within-site scatter (Table 1*b*). Because the magnetization was so stable, I used only stable endpoint data in the site means with a minimum of three samples per site. Alternating field demagnetization most often gave results similar to thermal demagnetization but occasionally failed to remove partial overprints, so the bulk of the collection was treated with stepwise thermal demagnetization.

Analysis of the results revealed three distinct magnetization types. Type 1 samples (n=14) exhibited just one component of magnetization with a steep downward direction, most commonly directed northwest (Figures 7*a* and 8*a*). Type 2R (n=11) samples revealed two components of magnetization: a secondary component directed north and steeply downward and a characteristic component directed southeast and upward (Figures 7*b* and 8*b*). Type 2N samples (n=9) had two distinct normal components of magnetization: a north and steeply downward directed secondary component followed by a downward directed, but scattered, characteristic component (Figures 7*c* and 8*c*).

Figure 7*c* also shows steps labeled NRM1 and NRM2. These measurements were made before and after several months storage in the very low-field room at UCSC. The viscous magnetization removed over that time was nearly parallel to the magnetization removed at temperatures from 100° to 300°C. This relationship suggests that the secondary magnetization observed in type 2R and 2N samples is a viscous magnetization. Viscous magnetizations persisting to such high temperatures are not predicted by single domain theory [*Néel*, 1955] but are not unknown [*Kent*, 1985].

Results. The type 1 and 2R remanent magnetizations are well grouped before tilt correction, while the 2N remanent magnetization is scattered both before and after tilt correction (Table 3 and Figure 8). The large amount of scatter in the 2N data precludes any meaningful interpretation. The source of

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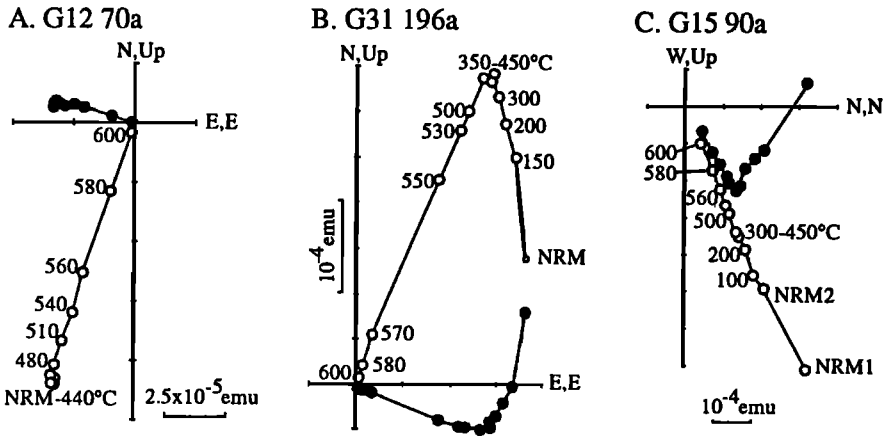


Fig. 7. Orthogonal vector diagrams of samples from Glacier Island; open symbols represent the horizontal component and closed symbols represent the vertical component. (*a*) Type 1 remanent magnetization, a northwest and steeply downward directed characteristic component with little or no secondary component. (*b*) Type 2R remanent magnetization. The characteristic component is directed southeast and up, with a north and down secondary component. (*c*) Type 2N remanent magnetization. Both the characteristic and secondary component have normal (downward) directions. The secondary component tends to be directed northward, while the characteristic component is scattered.

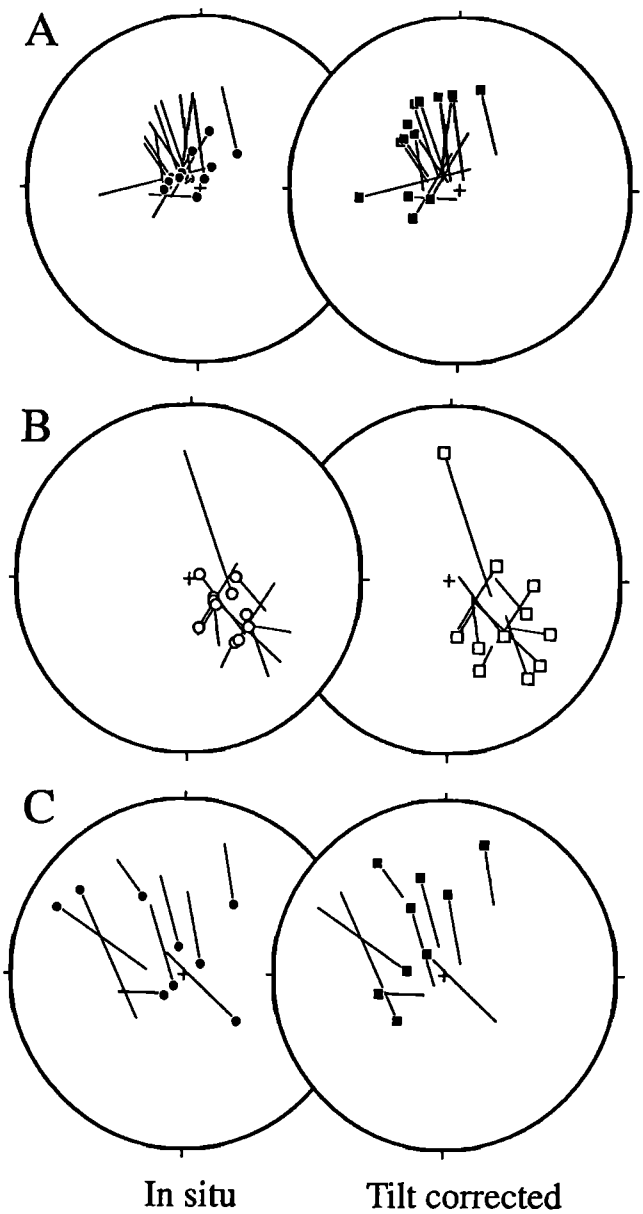


TABLE 3. Mean Paleomagnetic Directions, Glacier Island, In situ coordinates

	Declination	Inclination	*k*	a_{95}
Type 1 (n=14)				
In situ	330.1	81.8	44.8	6.0
Tilt corrected	321.3	57.3	10.8	12.6
Type 2R (n=11)				
In situ	129.5	-67.2	35.1	7.8
Tilt corrected	123.7	-54.2	6.1	20.2
Type 1&2R (n=25)				
In situ	316.0	75.6	30.3	5.3
Tilt corrected	313.5	56.3	8.2	10.8
Type 2N (n=9)				
In situ	324.6	73.0	4.7	26.6
Tilt corrected	328.2	61.2	5.4	22.7

k is dispersion parameter [*Fisher*, 1953]; a_{95} is alpha-95 [*Fisher*, 1953] 95% confidence limit.

this scatter may be explained by the large viscous component of magnetization. Rocks with such a "soft" magnetization probably acquired their remanence over a long period of time. Time which included hydrothermal alteration, deformation, apparent polar wander, and perhaps translation. This time-integrated magnetization could yield the result we observe: directions in the expected quadrant but too scattered for meaningful interpretation.

Type 1 and 2R remanent magnetizations fail a fold test (Table 3 and Figures 8*a* and 8*b*); after tilt correction the dispersion increases dramatically. The stereonet plots in Figure 8 show that there are large, significant variations in bedding attitude, which cannot be attributed to original topography. Therefore we can confidently say that these rocks acquired their remanence after deformation and the tilt-corrected mean directions are meaningless.

Fig. 8. Equal-area plots of site mean directions from Glacier Island. Closed (open) symbols are in the lower (upper) hemisphere; circles (squares) show in situ (tilt corrected) coordinates. The line attached to each point shows the effect of the tilt correction. (*a*) Type 1 remanence, (*b*) Type 2R remanence, and (*c*) Type 2N remanence. The type 1 and type 2R remanent magnetizations clearly fail a fold test and must have been acquired after deformation. The type 2N magnetization is scattered and of no use.

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Type 1 and 2R remanent magnetizations also fail a reversal test. When the type 2R directions are inverted through the origin, the type 1 and 2R mean directions (in situ) differ significantly (the hypothesis that type 1 and 2R directions share a common mean can be rejected with >95% confidence). The two most likely reasons for this are (1) the type 1 and 2R directions were acquired at different times and (2) secondary overprints were not completely removed during stepwise demagnetization. In the first case, if the type 1 and 2R remanence directions were acquired at different times, different directions could result from apparent polar wander, tilting of the ophiolite, or movement of the ophiolite to a substantially different latitude. In the second case, incomplete removal of a viscous overprint parallel to the present field direction (steeply down) would result in a characteristic direction which is too shallow for samples with a reversed (upward) polarity. In that case, one might expect to find that sites with higher dispersion (lower values of k, the *Fisher* [1953] precision parameter) would have shallower inclinations. However, there is no correlation between k and inclination for type 2R site mean directions. Furthermore, some sites have samples with characteristic directions of both polarities (e.g., site G20 has three of each, which are labeled G20a and G20b in Table 1*b*). This clearly shows that the normal and reversed remanent magnetizations were acquired at different times. Overall, the high quality of the data suggests there is no reason to suspect unresolved overprints, while there is every reason to expect tilting and/or displacement of the ophiolite. For these reasons, in my interpretation below, I treat the type 1 and 2R magnetizations as distinct and acquired at different times.

Table 2b reveals that type 1 samples are concentrated in sites G1-G10, which are located in close proximity to each other at the south end of Growler Bay (localities Ga-Gd, Figure 4) and close to the contact with the diabase dike complex. This suggests that there is some local geologic control on the magnetization type. However, type 1 magnetizations are also found in sites at the north end Growler Bay, as well as further west at Locality Gi, which has interbedded flysch, usually found in the upper portions of the pillow sequences. Thus there appears to be no direct correlation between magnetization type and either geographic location or stratigraphic position on Glacier Island.

Lithology also does not appear to exert a strong control over the magnetization type of the rocks (Table 4). Type 1 and 2N remanent magnetizations are split fairly evenly between dikes and pillows. The only correlation appears to be that the type 2R remanence resides principally in dikes. If the dikes intruded the pillows after they were tilted, then the type 2R magnetization may still be primary, despite the negative fold test. However, as discussed above, the dikes appear to grade into and feed the pillows. On a practical level, I cannot demonstrate that the type 2R magnetization is primary by any field stability test. Furthermore, whether the type 2R magnetization is primary or an overprint, the paleohorizontal reference has been lost and the data must be treated accordingly.

TABLE 4. Number of Sites of Each Remanence Type by Lithology

	Dikes	Pillows
Type 1	6	8
Type 2R	10	1
Type 2N	3	6
k<10	2	0
Unstable Remanent Magnetization	0	2
Total of all sites	21	17

Interpretation

The most conservative interpretation of the paleomagnetic results leads to the conclusion that the paleomagnetic directions from Knight Island are too scattered to be meaningful and the remanent magnetization from Glacier Island results from overprints. Since our paleohorizontal reference is then lost, tectonic interpretation is entirely speculative. In the case of Knight Island, I will take this conservative approach. On the other hand, the rocks at Glacier Island, while overprinted, display a coherence that warrants further analysis. In the discussion of Glacier Island results above, I argued for the presence of two distinct overprints, one normal, the other reversed. The interpretation I present below depends on this assumption. However, it should be noted that if the type 1 and 2R remanent magnetizations (with no tilt correction) are combined the resulting paleomagnetic pole is concordant with the North American apparent polar wander path (APWP) throughout the Paleogene.

Assuming two distinct overprints, the remanent magnetization at Glacier Island can be interpreted either in terms of tilting of blocks within the accretionary prism or displacement of the Chugach-Prince William terrane as a whole. To evaluate which interpretation is preferred, I first converted site mean directions to virtual (instantaneous) geomagnetic poles, then computed type 1 and type 2R mean paleomagnetic poles (Table 5). The mean paleomagnetic poles were compared to the early Tertiary North American APWP of *Diehl et al.* [1983]. The type 1 magnetization is concordant with the APWP and requires no tilting or displacement. The mean paleolatitude predicted by the type 1 magnetization is higher than the expected paleolatitude (if Glacier Island had stayed fixed to cratonal North America), but lies well within the error limits (Figure 2*a*).

The type 2R magnetization yields a pole which indicates either (1) approximately 30° of tilting down to the ESE or (2) 17°±10° of northward displacement with respect to cratonal North America, assuming that the magnetization was acquired at 55 Ma. The postulated tilting is counter to the observed bedding tilt, which primarily dips 30°-40° northwest (Figure 4). The displacement, on the other hand, agrees quite well with that observed at the Resurrection Peninsula (13°±9° [*Bol et al.*, 1992]). Furthermore, the remanent magnetization at Resurrection Peninsula is entirely reversed and southeast. Even the paleotrend of the sheeted dike complex is similar. For Glacier Island, the paleostrike of the dike complex is N51°E±32° (computed using the method of *Bol et al.* [1992]), while at Resurrection Peninsula it is N26°E±13°. The striking similarity of these two results does not appear to be

TABLE 5. Paleomagnetic Poles From Glacier Island

	Longitude °E	Latitude °N	k	a_{95}	Plat
Type 1	184.9	72.1	14.3	10.9	74.4
Type 2R	128.1	61.0	16.1	11.7	51.8
Type 1 and 2R	153.0	69.7	8.8	11.8	64.7

k is dispersion parameter [*Fisher*, 1953]; a_{95} is alpha-95 [*Fisher*, 1953] 95% confidence limit; Plat is paleolatitude.

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coincidental, but it does require that the Glacier Island ophiolite has remained essentially untilted since it acquired its overprint magnetization.

When was that magnetization acquired? Two lines of evidence appear to bracket the age of remanence acquisition between about 57 and 45 Ma: the age of the rocks and the arrival of the CPW in southwest Alaska. The age of the Glacier Island ophiolite is poorly known; however, similarities in lithology and sandstone composition provide a correlation with the Resurrection Peninsula [*Dumoulin*, 1988], which has a U-Pb date of 57±1 Ma [*Nelson et al.*, 1989]. This provides a reasonable upper limit on the age of the Glacier Island rocks and their magnetization.

A more tenuous constraint on the age of magnetization is provided by the paleomagnetism of rocks at the Resurrection Peninsula. The remanence there passes a fold test and appears to be primary [*Bol et al.*, 1992], that is, the magnetization was acquired when the rocks formed. Assuming that the remanence of the rocks at the Resurrection Peninsula and Glacier Island 2R are related, as suggested by their similarities, then the rocks at Glacier Island were probably overprinted soon after they formed, perhaps during accretion.

A weak lower bound on the age of overprinting is provided by the paleomagnetism of rocks in southwest Alaska and the Aleutian Islands. Studies there show that all terranes in southern Alaska were in place by about 45 Ma. The paleolatitude derived from the type 2R magnetization at Glacier Island indicates a large northward displacement, incompatible with acquisition of the type 2R remanence at 45 Ma.

In summary, by comparing the mean poles from the type 1 and 2R remanent magnetizations at Glacier Island to the early Tertiary North American APWP, we can draw conclusions about the nature and significance of the overprints. The type 2R magnetization, by its similarity to other paleomagnetic results in the CPW, appears to have been acquired early, probably during the process of accretion about 55 Ma, and at a position 17°±10° to the south. The type 1 magnetization is concordant with the North America APWP. If our interpretation of the type 2R magnetization is correct, then the type 1 magnetization was acquired after the CPW arrived in southern Alaska at approximately 45 Ma.

A final note and reminder: the remanent magnetization discussed above is an overprint; the interpretation depends on the assumptions that the rocks at Glacier Island acquired their magnetization soon after they formed and have not been appreciably tilted since they acquired their magnetization. I think that the similarities between the paleomagnetism of the Resurrection Peninsula, Ghost Rocks, and Glacier Island support these assumptions and warrant the interpretation. However, the evidence is circumstantial, and there is no independent evidence to confirm my assumptions.

Discussion

Paleomagnetic results from the CPW vary widely; this variability provided much of the motivation for the present study. The most reliable of the six studies (Figure 2), from the Ghost Rocks Formation [*Plumley et al.*, 1983] and the Resurrection Peninsula ophiolite [*Bol et al.*, 1992], have good age control, broad sampling, and detailed analysis, and their data pass stability tests. The previous reconnaissance result from Knight Island [*Hillhouse and Grommé*, 1977] appeared to record a primary magnetization and was the only one to yield a paleolatitude concordant with cratonal North America. However, this study demonstrates that the positive fold test in the prior Knight Island study was fortuitous and that the remanent magnetization recorded there is quite scattered and therefore of little value; the results should not be interpreted in terms of terrane displacement.

Results from Glacier Island, the Resurrection Peninsula, and the Ghost Rocks are concordant with coastwise transport using the full northward, coastwise displacement of the Kula plate (Figure 2*a*) [*Debiche et al.*, 1987]. The results of these studies carry important implications with regard to the nature of strike-slip faults along continental margins undergoing oblique convergence. First, since the discordance between the displacement histories of the Southern Alaska terrane and the CPW appears to be robust, it seems that faults within the forearc region accommodated most of the strike-slip motion generated by oblique convergence in the early Tertiary. This view contrasts with the more conventional wisdom that the most significant strike-slip faults are generated along the axis of the volcanic arc [e.g., *Fitch*, 1972]. Second, similarly, the displacement indicated by paleomagnetism provides a key example of large offsets which are plausible but have not been documented geologically. Third, the rapid poleward displacement of the CPW through Eocene time implies the continued presence of the Kula plate through Eocene time, as proposed by *Engebretson et al.* [1985] and *Lonsdale* [1988] but disputed by *Stock and Molnar* [1988].

Conclusions

Paleomagnetic results from Knight and Glacier islands yield overprint directions. At Knight Island, the directions are scattered and of no use for tectonic interpretation. The previous reconnaissance result of Hillhouse and Grommé, which implied no northward displacement with respect to cratonal North America, should be abandoned. Rocks from Glacier Island yielded three types of overprint magnetizations. Type 1 magnetizations are directed northwest and steeply down, type 2R magnetizations are directed southeast and moderate to steeply up, and type 2N magnetizations are directed downward but are scattered. Type 1 and 2R magnetizations appear to have been acquired at significantly different times; they are not antiparallel, and the difference between them does not appear to reflect unresolved partial overprints. The type 1 magnetization is concordant with the Paleogene North American reference pole. The type 2R magnetization is not concordant with the reference pole, and it is unlikely that it remagnetized during accretion or soon thereafter. The ophiolite has not been appreciably tilted since then. This interpretation explains the similarity of paleomagnetic results between Glacier Island and the Resurrection Peninsula in terms of both paleolatitude and paleotrend of dikes. This result supports the hypothesis of large displacement between the inboard Southern Alaska terrane and the outboard Chugach-Prince William terrane in Eocene time, suggested by other paleomagnetic and geologic studies.

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A. J. Bol, Columbine 18, Chautauqua Park, Boulder, CO 80302.

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